

APS110 – Lecture 22

See hand written notes for drawings.

Polymers, what we've talked about...

- Chain orientation
- Mer units & mer chemistry
- Chain length
- Cross-linking
- Crystallinity

Crystallinity → how can we control crystallinity of polymers

- Use simple, relatively small repeating units; linear polymers
- Linear polymers
 - In reality, most polymers have a little more complexity to them than just squiggles on a page (like we've been drawing)

Branching (feature of molecule structure)

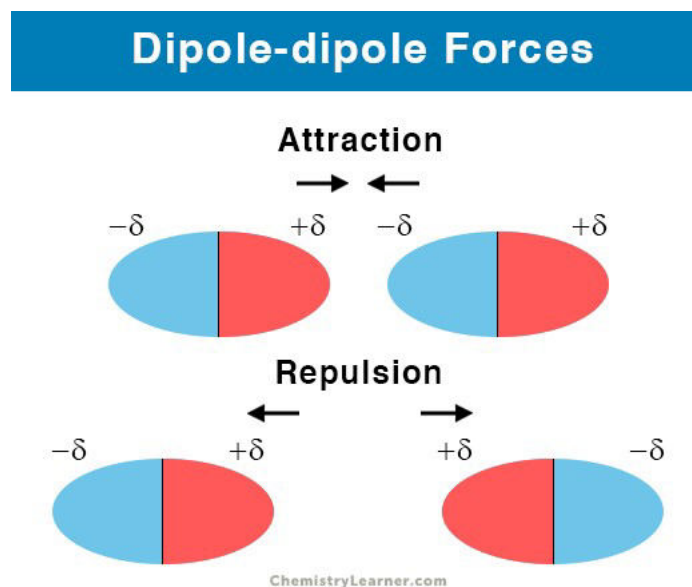
- Branches off of main chain
- These branches could have the same mer unit as main chain (same chemistry), or can be different
- Held to the main chain by a **strong, covalent bond**
- Ex. LDPE & HDPE
 - HDPE has fewer branches and they are smaller
 - This means the molecules can get closer to each other and pack tighter, which means there is a higher degree of crystallinity → higher strength

Reducing branching ← one way to control crystallinity, at the molecular level

We can also control crystallinity by considering 3D shape of molecule

- **R** – denotes an unknown, or undescribed functional group
 - Could be *anything* (methyl group, ring structure, etc.) that is **NOT high energy**
- All functional groups on the same side, as per orientation of molecule = **ISOTACTIC**
 - Regular, repeating structure
 - Anything regular like this **will favour crystallinity**
 - Since all the functional groups are on the same side, the molecules can pack close together and the functional groups won't bump into each other
- **ATACTIC** = **random** arrangement of functional groups
 - Most unlikely to crystallize
- **SYNDIOTACTIC** = functional groups are **alternating**
 - Out-in-out-in-out-in (L – R – L – R – L – R)
 - Some *intermediate* ability to crystallize, between atactic and isotactic

- When the molecules can pack together tightly (i.e. crystallinity) there are more weak secondary bonds that hold molecules together and thus the strength of the material increases
- What if we can control the magnitude (strength) of that weak bond?
 - INTERMOLECULAR bonds
- Ex.
 - Cl is highly electronegative (attracting electrons)
 - When involved in the mer unit with C and H, the Cl will pull the electrons towards itself and will thus develop a **partial negative charge (δ^-)**
- Secondary bonds are INTERmolecular bonds – bonds between molecules
- Secondary bonds are MUCH weaker than primary, intramolecular bonds
- Dipole-dipole bond



- If we set up more electronegative functional groups, we can elicit stronger secondary bonding

Temperature is very important to polymers, as well as the **rate** in which we load them

→ we need to consider the influence of TIME and TEMPERATURE

- Ex. PMMA (acrylic, plexiglass)
 - How can there be such a big difference for 20°C change in temperature?
 - Balance between binding energy and the kinetic, thermal energy
 - Because polymer properties are influenced so much by secondary bonds, and secondary bonds (bond strength) are similar in energy to thermal energy, small changes to thermal energy (temperature) have large impact on properties of polymers
 - We can overcome the bond strength more when the molecules have more kinetic energy (increased temperature)

- o At 0°C we have removed enough kinetic energy that the secondary interactions win → thus we get brittle fracture of the polymer (since bond strength is high)
- o As you increase the temperature, there is a battle of secondary bonds and kinetic energy – these are on the same order of magnitude, so small changes in temperature make a big differences
 - Moderate temperature changes for metals and ceramics don't make much difference on the whole
- **Time → polymers will deform as a function of time**
 - o A small load applied to a polymer over a long period of time can generate a change in strain
 - This does not happen with metals or ceramics
 - o Consider a mass applied over time
 - Think about a coffee table on a carpet over time
 - As time passes, the polymer fibres in the carpet become flattened under the feet of the table
 - Depending on how long the table has been there (i.e. years) even after you move the table the carpet will remain compressed where the table sat
 - The polymer fibers have accumulated strain over time under the heavy weight